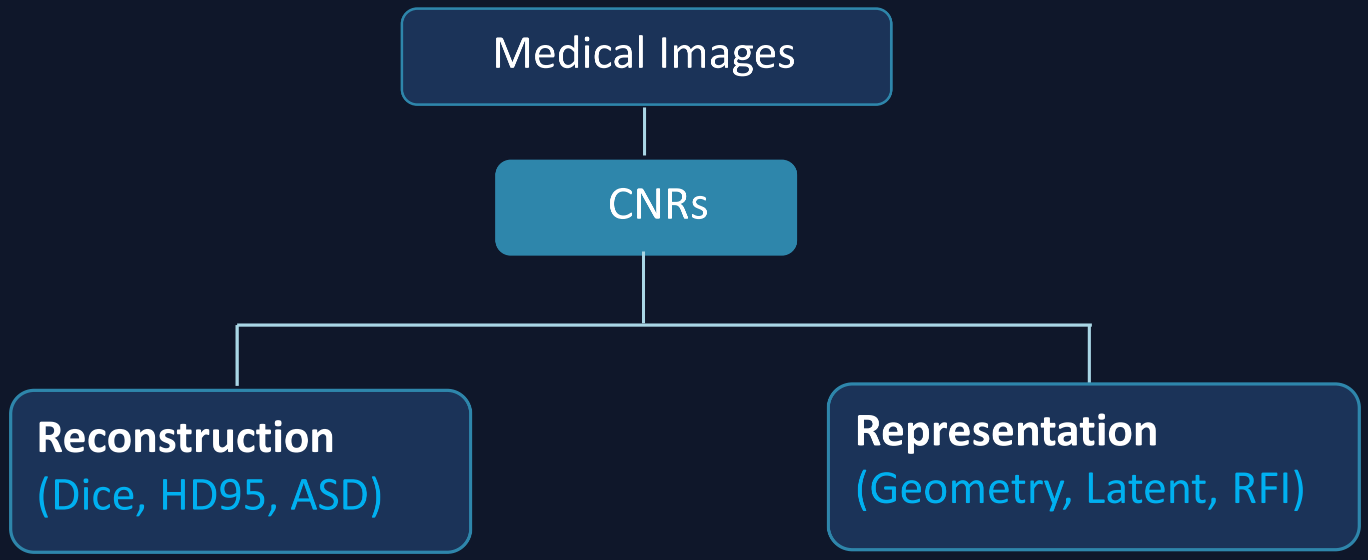
Beyond Reconstruction: Measuring What Metrics Missing



Can reconstruction accuracy faithfully characterize a continuous neural representation?

Pronab Kumar Paul, Department of Information and Communication Engineering, University of Rajshahi

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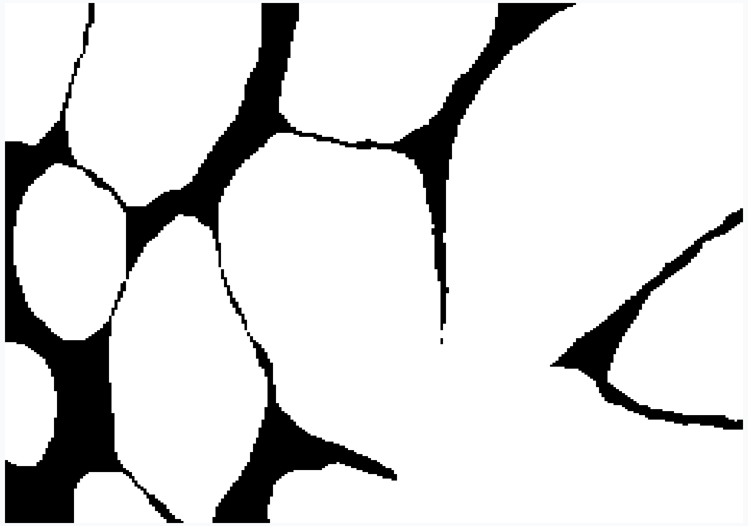
THE QUESTION MOTIVATING THIS WORK

Are High Dice Scores Enough?

Can identical reconstruction imply identical learned representations?



Ground Truth

Original Target Structure

Model A (Dice=0.994)

Model B (Dice=0.985)

Why Reconstruction Can Be Misleading

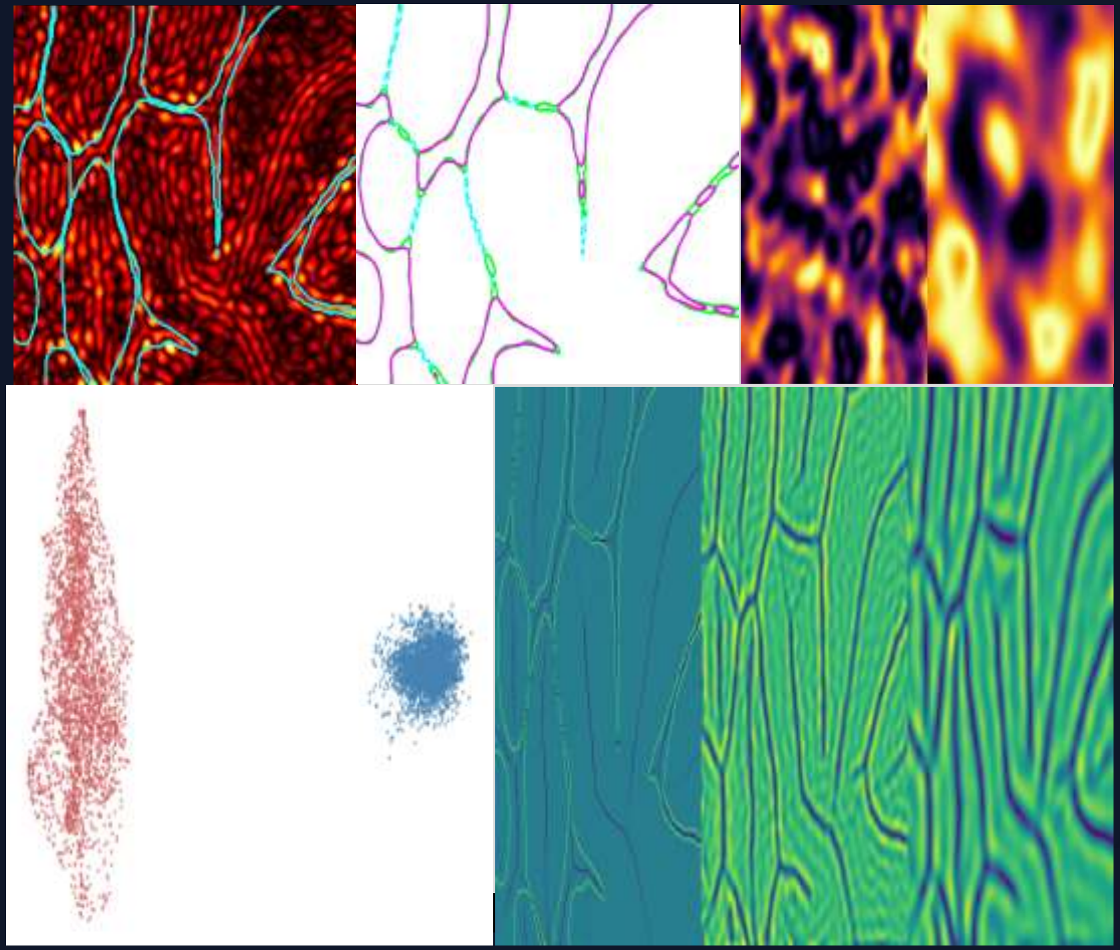
→ Equivalent reconstruction performance

→ Different learned representations

→ Different representational fidelity

→ Conventional reconstruction metrics cannot detect these differences

→ Representation-aware evaluation is needed



Equivalent Reconstructions Can Hide Different Representations.

Equivalent Reconstruction ≠ Equivalent Representation

Equivalent reconstructions can hide substantial differences in representational fidelity.

Representational fidelity: The extent to which anatomically meaningful information is preserved within the learned representation.

Research Questions

Characterizatio n

How do learned representations differ despite equivalent reconstruction?

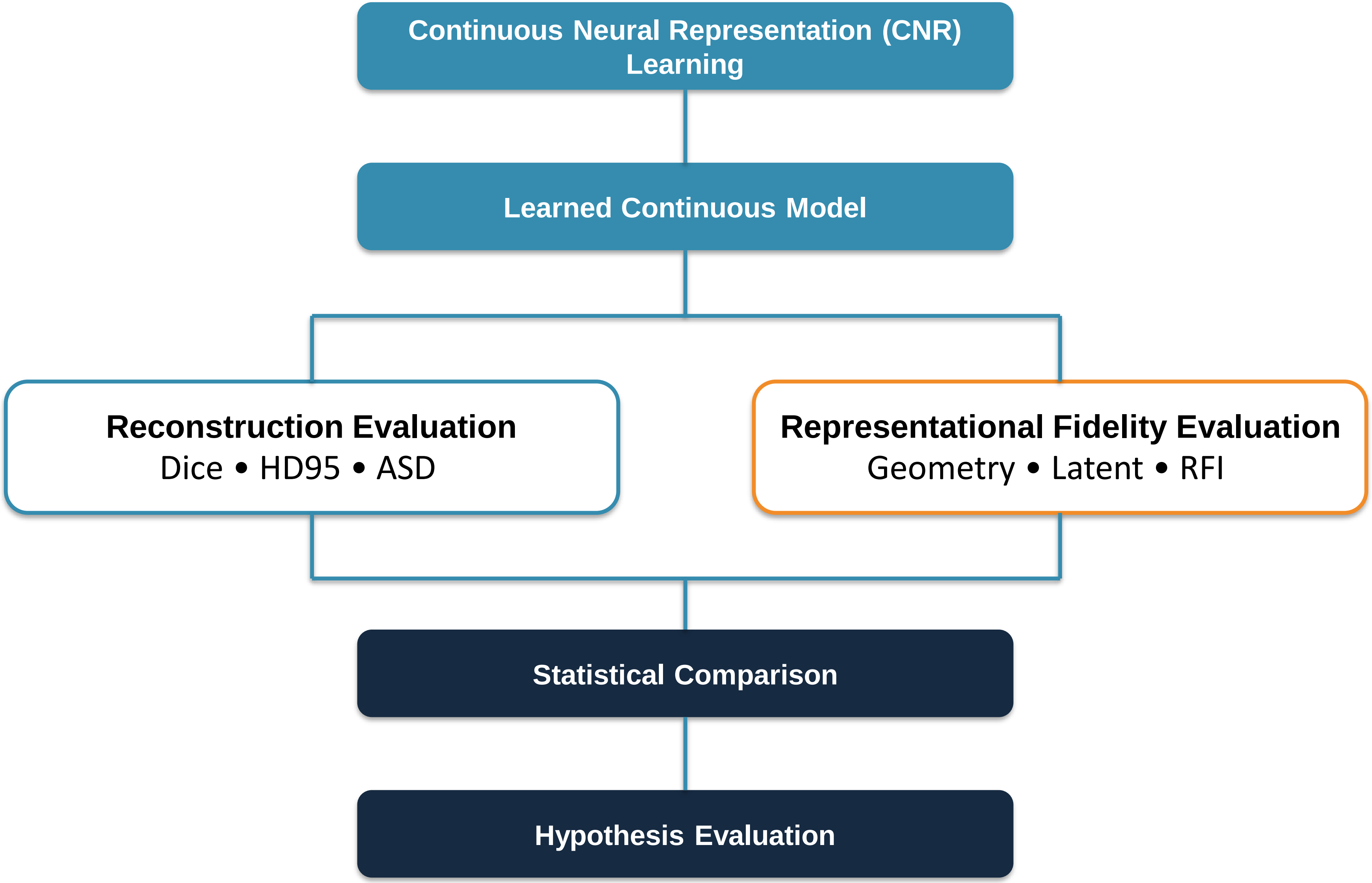
Quantification

Can a composite Representational Fidelity Index (RFI) quantify these differences?

Generalization

Do these findings generalize across datasets and model architectures?

Proposed Evaluation Framework



How is RFI Computed?

Representational Fidelity Index (RFI) integrates geometric fidelity and latent representation quality into one score.

Geometry Index (GeoIndex)

- Gradient Error
- Curvature Error
- Topology Error (Euler Characteristic)

Latent Index (LatentIndex)

- Effective Rank
- Spectral Entropy
- Participation Ratio

Standardize All Metrics (Z-score)

```
RFI = (GeoIndex + LatentIndex) / 2
```

CKA reported separately to validate latent-space divergence.

Experimental Validation

DRIVE Dataset

Retinal Vessels

→ 20 retinal images

→ 3 random seeds per model

→ Matched reconstruction pair identified

→ Wilcoxon + Holm

GLAS Dataset

Gland Structures

→ 20 histological images

→ 2 random seeds per model

→ **Statistical Test:** TOST equivalence

→ Equivalence bound ±0.02

Both datasets produced statistically equivalent reconstruction while revealing substantial representational differences.

DRIVE Dataset: Equivalent Reconstruction

~0.0

P-VALUE DIFF (NS)

Wilcoxon + Holm: all adjusted $p > 0.05$

Metric	SIREN-128	SIREN-112	Result
Dice ↑	0.722	0.719	✓ ns
HD95 ↓	5.81	5.83	✓ ns
ASD ↓	0.610	0.616	✓ ns

Standard reconstruction metrics indicate that the two models are equivalent.

DRIVE: Representational Fidelity

Representation Metrics

Metric	SIREN-128	SIREN-112	Result
RFI ↑	−0.234	−0.145	**$p < 0.0001$**
LatentIndex ↑	−0.286	−0.150	**$p < 0.0001$**
Effective Rank ↑	84.84	74.02	**Significant**

Effective Rank: **84.84 → 74.02** *(≈13% lower)*

LatentIndex: **−0.286 → −0.150** *(large effect)*

RFI: **−0.234 → −0.145** *(Cohen's d = −2.65)*

Geometry Metrics

Metric	SIREN-128	SIREN-112	Result
Gradient Error ↓	30.44°	30.55°	**$p = 0.040$**
Curvature Error ↓	0.955	0.960	**$p = 0.033$**
Vessel Width Error ↓	2.60	2.67	**trend ($p = 0.097$)**

Despite equivalent reconstruction, the learned representations differ significantly across latent and geometric analyses.

GLAS Dataset: Equivalent Reconstruction

< 0.0001

TOST P-VALUE

All pairs equivalent

Reconstruction Metrics

Model	Dice ↑	HD95 ↓	ASD ↓
SIREN-128	0.995	1.85	0.151
SIREN-128 + Eikonal	0.995	2.15	0.184
SIREN-128 + Jacobian	0.995	2.08	0.177
SIREN-128 LowOmega (ω_0=15)	0.986	5.43	0.591

Standard reconstruction metrics suggest that all four models are equivalent.

GLAS: Representational Fidelity

Representation Metrics

Model	RFI ↓	Effective Rank ↑	Participation Ratio ↑
SIREN-128	−0.303	74.44	51.73
SIREN-128 + Eikonal	−0.296	74.08	51.41
SIREN-128 + Jacobian	−0.322	74.29	51.58
SIREN-128 LowOmega (ω_0=15)	**+0.920**	**43.54**	**28.46**

Structural Differences

Metric	SIREN-128	LowOmega
Gradient Error	10.58°	**12.45°**
Curvature Error	1.010	**1.042**
GeoIndex	−0.131	**0.462**
LatentIndex	−0.475	**1.379**

✓ 41% reduction in Effective Rank **✓ ≈45% reduction in Participation Ratio**

✓ RFI shifted from −0.303 to +0.920 **✓ Non-overlapping 95% confidence intervals**

Equivalent reconstruction concealed substantial degradation in geometric fidelity and latent representation quality.

Consistency Across Datasets

Dataset	Reconstruction	Representation
DRIVE (Retinal Vessels)	Equivalent	Significantly Different
GLAS (Gland Structures)	Equivalent	Significantly Different

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datasets confirm representation divergence

Equivalent reconstruction does not imply equivalent learned representations.

The phenomenon replicates across anatomically distinct datasets.

Why Does This Happen?

Model A

Reduced feature diversity

↓

Fewer active feature directions

↓

Lower effective rank (observed)

↓

Reduced latent representation richness (observed)

↓

Equivalent reconstruction despite reduced representational fidelity

Model B

Greater feature diversity

↓

More active feature directions

↓

Higher effective rank (observed)

↓

Richer latent representation (observed)

↓

Equivalent reconstruction with higher representational fidelity

Equivalent reconstruction may arise from fundamentally different internal representations because reconstruction evaluates the final output, not how anatomical information is organized within the learned feature space.

CONCLUSION

Current Paradigm

Model quality inferred from reconstruction performance

➔

Proposed Paradigm

Model quality evaluated through reconstruction and representational fidelity

Findings

- ✓ Equivalent reconstruction does not imply equivalent learned representations.
- ✓ RFI quantifies representational fidelity beyond reconstruction metrics.
- ✓ The phenomenon was consistently observed across DRIVE and GLAS datasets.
- ✓ Representation-aware evaluation provides a more complete assessment of continuous neural representations.

Accurate reconstruction does not necessarily imply faithful learned representations.

Toward representation-aware evaluation for more reliable medical AI.

Questions?

"Accurate reconstruction does not necessarily imply faithful learned representations."

Pronab Kumar Paul

University of Rajshahi

pronabpaul77@gmail.com

github.com/pronabpaul/revealing-representational-fidelity

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